

TTT4295 - Acoustic Signal Processing

Autumn 2026

Course plan

Lecture time: Mon. 1215 - 1400
Tue. 1415 - 1600

Exercise time: Fri. 0815 - 1000 (**starting week 35**)

Place: (Lectures) [R40 Realfagbygget](#)
(Exer.) [429B Hovedbygningen](#)

Teachers: SM Sara Martin (course coordinator)
UPS Peter Svensson
GD Guillaume Dutilleux

Coordinator for the course: Sara Martin - sara.martin@ntnu.no

Teaching assistant: Erik Lagerqvist Dudek - erikld@stud.ntnu.no

Syllabus:

Course compendium: P. Svensson - TTT4295 Acoustic Signal Processing
F. F. Li, T. J. Cox - *Digital Signal Processing in Audio and Acoustical Engineering*
U. Zölzer - [Digital Audio Effects](#)

Week	Date	Content	Lecturer
34	Mon.17.08 & Tue. 18.08	Acoustic Signals and Systems <i>Compendium & Chp 1. Cox</i>	UPS
35	Mon.24.08	Acoustic Signals and Systems (cont) + Intro Assig. 1 <i>Compendium & Chp 1. Cox</i>	UPS
	Tue. 25.08	Sampling and Fourier <i>Chp 2. Cox</i>	
36	Mon 31.08	Electroacoustics: Loudspeakers & Microphones <i>Compendium</i>	UPS
	Tue. 1.09	Responses of Audio Systems <i>Compendium</i>	
	Fri. 4.9	Deadline for exercise 1	
37	Mon 7.09	Binaural Hearing <i>Compendium & Chp 6.1 Cox</i>	UPS
	Tue. 8.09	Binaural Hearing (directional) <i>Compendium & Chp 6. Cox</i>	
38	Mon 14.09	Microphone Arrays <i>Compendium & Chp 6.2 Cox</i>	UPS
	Tue. 15.09	Microphone Arrays <i>Compendium & Chp 6.2 Cox</i>	
	Fri.18.9	Deadline for Assignment 1 (small deliverable)	
39	Mon 21.09	Microphone Arrays <i>Compendium & Chp 6.2 Cox</i>	UPS
	Tue. 22.09	Auralization and Virtual Acoustics + Intro Assign. 2	
	Fri. 25.9	Deadline for exercise 2	
40	Mon.28.09	Auralization and Virtual Acoustics <i>Compendium</i>	UPS
	Tue. 29.09	Active Noise Control and Adaptive filters <i>Chp 7. Cox (except 7.4)</i>	
41	Mon. 5.10	Active Noise Control and Adaptive filters <i>Chp 7. Cox (except 7.4)</i>	UPS
	Tue. 6.10	Backup session	
	Fri. 9.10	Deadline for Assignment 2 (small deliverable)	
42	Mon. 12.10 & Tue. 13.10	Music Synthesis	SM
43	Mon. 19.10	Physical modeling	SM

	Tue. 20.10	Guest lecture	
	Fri. 23.10	Deadline for exercise 3	
44	Mon. 26.10	Speech synthesis	SM
	Tue. 27.10	Guest lecture	
	Fri. 30.10	Deadline for Assignment 3 (small deliverable)	
45	Mon. 2.11 & Tue. 3.11	Advanced Acoustic Signal Representation	GD
	Fri. 7.11	Deadline for exercise 4	
46	Mon. 9.11 & Tue. 10.11	Machine Learning principles	GD
47	Mon. 16.11 & Tue. 17.11	Machine Learning principles	GD

ASSIGNMENTS & EXERCICES

There are three mandatory assignments and four mandatory exercises, with deadlines as listed in the table above.

The assignments are larger tasks, each requiring a small deliverable. You must complete all three. After the last assignment has been completed, one assignment will be selected at random for an oral presentation, and another for a written full report — each counting **15%** of your final grade. **The report must be submitted by the day of the exam, 30th of November.**

While you may collaborate in pairs or groups, each student must write their own report and give an individual presentation. Assignments typically involve some form of experiment.

The exercises are shorter tasks and do not require formal reports. You are expected to attempt all mandatory problems and submit your solutions.

EXAM

Written exam on the 30th of November 2026 at 15h00